```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score

# Generate dataset
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([2, 4, 5, 8, 10, 12, 14, 16, 18, 20])

# Display dataset
print("X values:")
print(X)

print("\ny values:")
print(y)
```

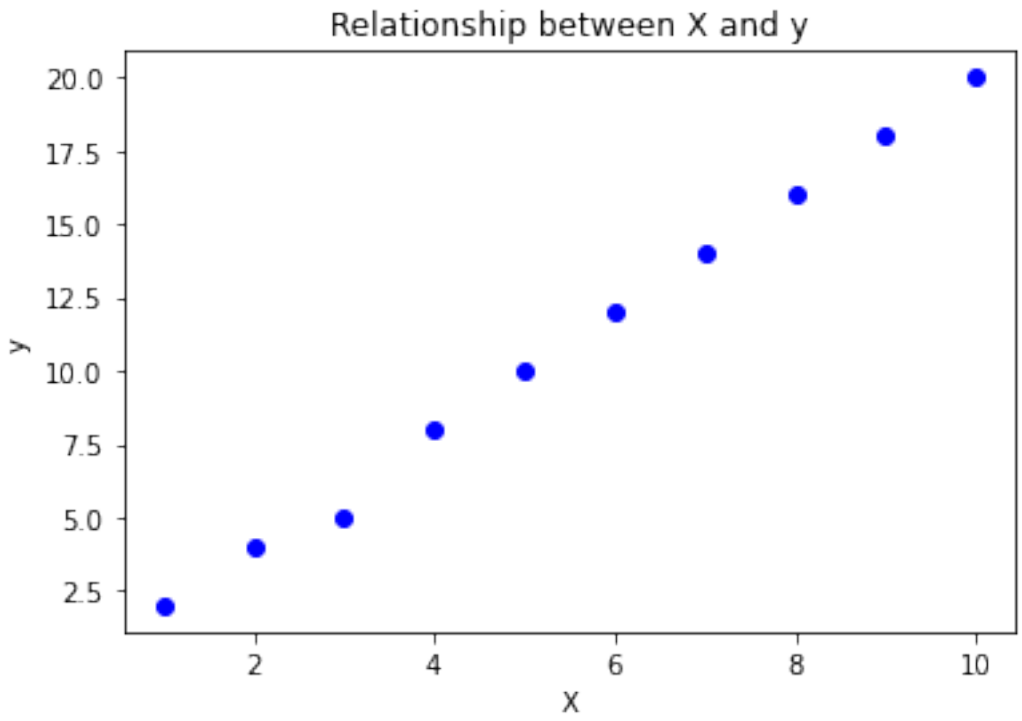
```
X values:
[[ 1]
 [ 2]
 [ 3]
 [ 4]
 [ 5]
 [ 6]
 [ 7]
 [ 8]
 [ 9]
 [10]]

y values:
[ 2  4  5  8 10 12 14 16 18 20]
```

```
#Visualize the relationship between X and y
plt.scatter(X, y, color='blue')

plt.xlabel('X')
plt.ylabel('y')
plt.title('Relationship between X and y')

plt.show()
```




```
#Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training data:")
print(X_train)

print("\nTesting data:")
print(X_test)
```

```
Training data:
[[ 6]
 [ 1]
 [ 8]
 [ 3]
 [10]
 [ 5]
 [ 4]
 [ 7]]

Testing data:
[[9]
 [2]]
```

```
#Create and train the Linear Regression model
model = LinearRegression()
```

```
# Train the model
model.fit(X_train, y_train)

print("Model trained successfully!")
```

```
Model trained successfully!
```

```
#Make predictions
y_pred = model.predict(X_test)

print("Actual values:", list(y_test))
print("Predicted values:", list(y_pred))
```

```
Actual values: [18, 4]
Predicted values: [18.025862068965516, 3.724137931034482]
```

```
#Evaluate the model using MAE, MSE, RMSE and R²
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root Mean Squared Error (RMSE):", rmse)
print("R² Score:", r2)
```

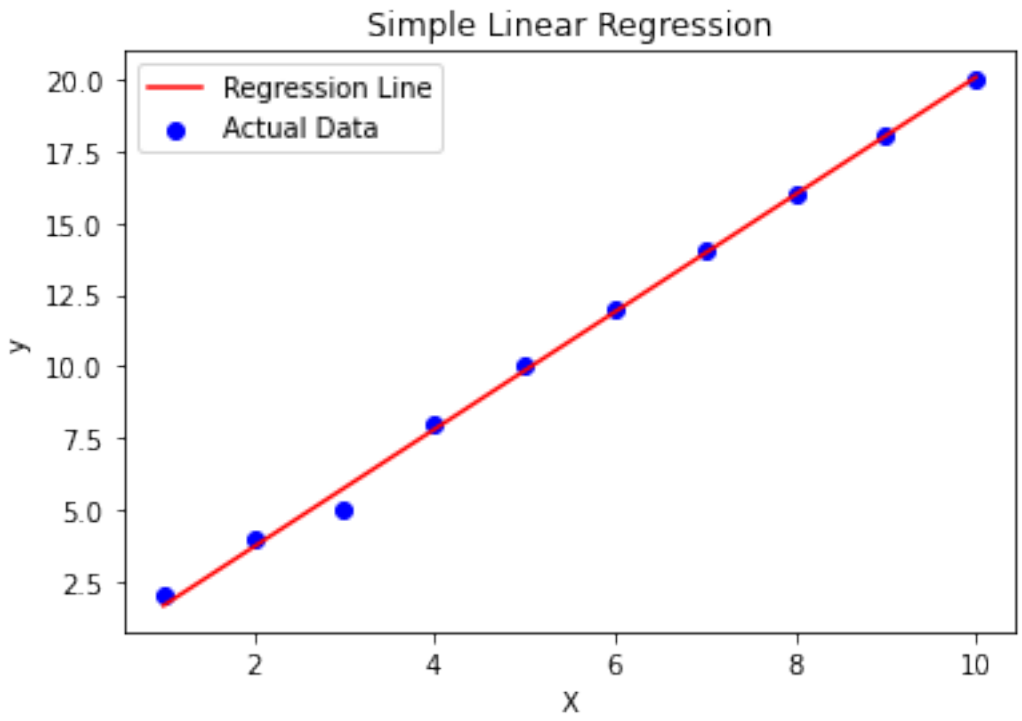
```
Mean Absolute Error (MAE): 0.1508620689655169
Mean Squared Error (MSE): 0.0383843638525566
Root Mean Squared Error (RMSE): 0.19591927892006086
R² Score: 0.9992166456356621
```

```
#Plot the regression line with actual data points
plt.scatter(X, y, color='blue', label='Actual Data')

# Regression line
plt.plot(X, model.predict(X), color='red', label='Regression Line')

plt.xlabel('X')
plt.ylabel('y')
plt.title('Simple Linear Regression')
plt.legend()

plt.show()
```




```
#Predict a new value
new_experience = [[11]]

prediction = model.predict(new_experience)

print("Predicted Salary:", prediction[0])
```

```
Predicted Salary: 22.112068965517242
```